

Airbnb Listings Analysis

1. Project Overview

This project analyses the Airbnb listings dataset to understand listing characteristics, pricing patterns, room-type popularity, neighbourhood distribution, host verification, booking availability, review ratings, minimum stay requirements, and relationships between different listing variables.

The objective of the analysis is to identify important patterns in Airbnb listings and understand how factors such as room type, location, price, reviews, host verification, instant booking, and availability influence the overall characteristics of the listings. The analysis was performed using Python, Pandas, Matplotlib, and Seaborn.

2. Dataset Summary

- **Rows:** 102,599
- **Columns:** 26 before data cleaning
- **Dataset:** Airbnb_Open_Data
- **Main Features:** Listing ID, listing name, host details, neighbourhood, latitude, longitude, country, room type, construction year, price, service fee, minimum nights, number of reviews, last review, reviews per month, review rating, host listing count, availability, and house rules.
- After removing records with missing listing and host names and duplicate records, the dataset contained 101,410 records. - The dataset contains missing values in several numerical and categorical fields, which were handled during the data-cleaning process.

3. Exploratory Data Analysis using SQL

Exploratory Data Analysis (EDA) was performed using Python to understand the structure, quality, and characteristics of the Airbnb listings dataset.

- **Data Loading:** Imported the Airbnb dataset using Pandas and examined the first few records.
- **Initial Exploration:** Used `df.columns`, `df.shape`, `df.dtypes`, `df.info()`, and `df.describe()` to understand the dataset structure and statistical characteristics.
- **Missing Data Handling:** Missing values were identified using `df.isnull().sum()`. Several categorical and numerical columns contained missing values.
- **Date Conversion:** The last review column was converted into a datetime format for consistent analysis.
- **Record Cleaning:** Rows with missing listing names and host names were removed.
- **Duplicate Records:** Duplicate records were identified, with 539 duplicates found and removed from the dataset.

- **Price Conversion:** The price and service fee columns originally contained currency symbols and were converted into numerical values for analysis.
- **Unnecessary Column Removal:** The license column was removed because it contained almost entirely missing values.
- **Missing Value Imputation:** Missing categorical values were replaced with "Unknown" or appropriate values, while numerical missing values such as price and service fee were handled using median values.
- **Data Standardization:** Incorrect neighbourhood group values such as brookln and manhatan were corrected to Brooklyn and Manhattan.
- **Outlier Handling:** Minimum nights and availability values were constrained to reasonable ranges. Minimum nights were limited between 1 and 365 days, while availability was limited between 0 and 365 days.
- **Final Data Validation:** The final null-value check showed that the major dataset fields had been cleaned and contained no remaining missing values.

4. Data Analysis and Key Business Insights

The cleaned dataset was analysed to understand important patterns in Airbnb listings.

- **Room Type Popularity:** Entire home/apt listings represent the largest share of the dataset at 52.39%, followed by Private rooms at 45.33%. Shared rooms account for 2.17%, while Hotel rooms represent only 0.11% of listings.
- **Price Distribution:** The distribution of listing prices was analysed using a histogram with a KDE curve to understand the overall spread of Airbnb prices.
- **Price by Room Type:** Average and median prices were compared across different room types. Hotel rooms had the highest average price at approximately \$666.39, followed by Shared rooms at \$634.51, Entire home/apt at \$625.38, and Private rooms at \$624.84.
- **Listings by Neighbourhood Group:** The number of Airbnb listings was compared across neighbourhood groups to understand the geographic distribution of listings.
- **Neighbourhood Pricing:** Manhattan had 43,279 listings, Brooklyn had 41,364 listings, Queens had 13,120 listings, Bronx had 2,678 listings, and Staten Island had 943 listings. The average prices were relatively close across the major neighbourhood groups, ranging from approximately \$622.83 in Manhattan to \$629.97 in Queens.
- **Price by Room Type Visualization:** A box plot was used to compare the distribution of prices across room types while limiting the visualization to listings priced at or below \$1,000.
- **Review Rating Analysis:** The distribution of review ratings was analysed to understand the quality ratings across Airbnb listings. Review ratings were also compared across different room types.

- **Availability Analysis:** Listing availability across 365 days was analysed by room type using box plots to understand differences in availability patterns.
- **Host Verification Analysis:** Host verification status was analysed to understand the distribution of verified and unverified hosts. Price differences by host verification status were also explored.
- **Instant Booking Analysis:** Instant booking availability was analysed to compare listings that allow instant booking with those that do not. The dataset contains 50,444 instant-bookable listings and 50,966 listings that are not instant-bookable. The average prices for both groups were almost identical, at approximately \$625.37 and \$625.39 respectively.
- **Minimum Stay Analysis:** The distribution of minimum stay requirements was analysed, focusing on listings with minimum stay requirements of 30 days or less.
- **Service Fee Analysis:** The service fee was compared with listing price to examine the relationship between the two values. The calculated fee percentage had a mean of approximately 0.20, indicating that the service fee is generally close to 20% of the listing price, although some records contain unusual values.
- **Correlation Analysis:** A correlation matrix was created for numerical variables including price, service fee, minimum nights, number of reviews, reviews per month, review rating, host listing count, and availability. A heatmap was used to visually examine relationships between these variables.

5. Data Visualizations

Several visualizations were created using Matplotlib and Seaborn to communicate the findings effectively.

The analysis includes:

- Price Distribution Histogram
- Room Type Popularity Count Plot
- Airbnb Listings by Neighbourhood Group
- Price Distribution by Room Type Box Plot
- Median Airbnb Price by Neighbourhood Group
- Review Rating Distribution
- Review Ratings by Room Type
- Availability by Room Type
- Host Verification Status
- Price by Host Verification Status
- Instant Booking Availability
- Distribution of Minimum Stay Requirements
- Correlation Heatmap of Airbnb Listing Variables

These visualizations provide a clear understanding of pricing, room types, location, reviews, availability, host characteristics, booking options, and relationships among numerical variables.

6. Business Recommendations

- **Focus on High-Demand Room Types:** Since Entire home/apt and Private room listings account for the majority of listings, hosts can focus on maintaining competitive pricing and improving the quality of these popular accommodation types.
- **Optimize Pricing:** Hotel rooms have the highest average price among the room types analysed. Hosts can use room-type and neighbourhood pricing patterns to develop competitive pricing strategies.
- **Improve Listing Visibility:** Since Manhattan and Brooklyn contain the largest number of listings, hosts in these areas should focus on strong listing descriptions, competitive pricing, high-quality images, and positive reviews to differentiate their properties.
- **Encourage Positive Reviews:** Review ratings are an important part of the listing analysis. Hosts should maintain service quality and guest experience to encourage better ratings and improve listing attractiveness.
- **Use Instant Booking Strategically:** Instant-bookable listings and non-instant-bookable listings have very similar average prices. Therefore, instant booking can be considered primarily as a convenience and customer-experience feature rather than simply a pricing strategy.
- **Manage Minimum Stay Requirements:** Hosts should set minimum stay requirements based on demand patterns. Excessively high minimum-night requirements may reduce booking opportunities, while reasonable requirements can help improve occupancy.
- **Monitor Availability:** Availability patterns should be regularly monitored across room types and neighbourhoods to identify opportunities for improving occupancy and revenue.
- **Review Service Fee Consistency:** The analysis indicates that the service fee is generally close to 20% of the listing price, but unusual fee percentages are present. These records should be reviewed to ensure pricing and fee information is consistent.
- **Maintain Data Quality:** Regular data validation should be performed to identify incorrect values, duplicate records, missing information, and inconsistent categorical values so that future analysis remains reliable.

7. Conclusion

The Airbnb Listings Analysis provides an overall understanding of listing distribution, room types, pricing, neighbourhood characteristics, reviews, availability, host verification, instant booking, and minimum stay requirements.

The analysis shows that Entire home/apt listings form the largest proportion of the dataset, while Manhattan and Brooklyn contain the majority of listings. Pricing is relatively similar across the major neighbourhood groups, whereas Hotel rooms have the highest average price among the analysed room types. The analysis also highlights the importance of data cleaning, particularly for missing values, duplicate records, incorrect categorical values, currency-formatted columns, and unrealistic numerical values.

Overall, the Python-based EDA provides useful insights that can help Airbnb hosts and platform stakeholders understand market patterns, improve pricing decisions, enhance listing quality, and make more informed business decisions.